Quiz 4- Solution

(1) What is a REST interface?

Answer: REST interfaces provide Web services by using Web protocols such as HTTP. They have features like scalability, maintainability, help multiple application communication built on various programming languages, etc.
REST-based Web service implementation defines the method of accessing various resources that are required by the client.

(2) List the features of REST-based Web services.

Answer: Every REST-based service has the following features:

- It works based on the client-server communication model.
- Commonly uses the HTTP protocol for performing functions like retrieving resources from the web server, execution of any query, etc.
- The client-server communication is performed through 'messaging'.
- Resources available on the server are addressed through URIs.
- It is stateless in that every client-server request-response is independent of each other with complete assurance of providing required information.
- It works based on the uniform interface (HTTP verbs).
- It uses the concept of caching.

(3) Describe a Web resource in the context of the REST interface.

Answer: A web resource can be any data object of type image, HTML file, text data, and video. There are varieties of resource representation formats and some most common ones are- JSON, YAML, XML, and HTML.
https://www.softwaretestinghelp.com/restful-web-services-interview-question/

(4) Explain how the OAuth 2.0 works?

Answer: OAuth 2.0 is used for the authorization of secure access to protected data stored on a server, and the request workflow is performed as the below:

- When a user attempts to access a resource on a secure server, the user's application initiates the OAuth 2.0 authorization process.
- The request is directed to the authorization server with a scope specifying actions to retrieve, insert, update, or delete when authenticated.
- The authorization server returns an access token to the client application to access those resources.

(5) What is the Internet of Things? Explain the role of mobile applications in IoT.

Answer: IoT is a system of internet-connected objects that collect and transfer data over a wired as well as wireless network and brings endless possibilities without human intervention. It collects data through sensors to the cloud, processes it and then decide

to perform an action. Mobile applications play a significant role and used as an interface for users to interact with IoT-enabled physical devices.

References:

- https://simpleprogrammer.com/oauth-and-rest-in-android-part-1/
- https://en.wikipedia.org/wiki/Representational_state_transfer